## Analytical Approach

- **Reliable insights start with clean data.**
  Duplicates, missing values, and outliers can easily distort results, so careful preprocessing is essential.

- **Quantify effects, not just detect them.**
  Instead of “effect vs no effect”, estimate **how strong the effect is**, what impact it could have, and **how confident we are** in the estimate.

- **Prefer simple insights that drive decisions.**
  A straightforward, actionable finding is often more valuable than a complex but non-monetizable pattern in the data.

# Insight 1

**Publishing day and video format clearly affect performance.**

- **But by how much?**
- **How many additional views can we get by publishing on the right day?**
- **And how much of a boost does the format actually give?**

## Data

- YouTube video performance dataset
- Features: publishing weekday, video format
- Target: views during the first 7 days after publication
- Filtered: non-zero views, missing values removed

## Hierarchical Bayesian Model

**Other factors: channel audience, topic, external events, etc.**

$$y_i = \log(1 + \text{views}_{i,7d})$$

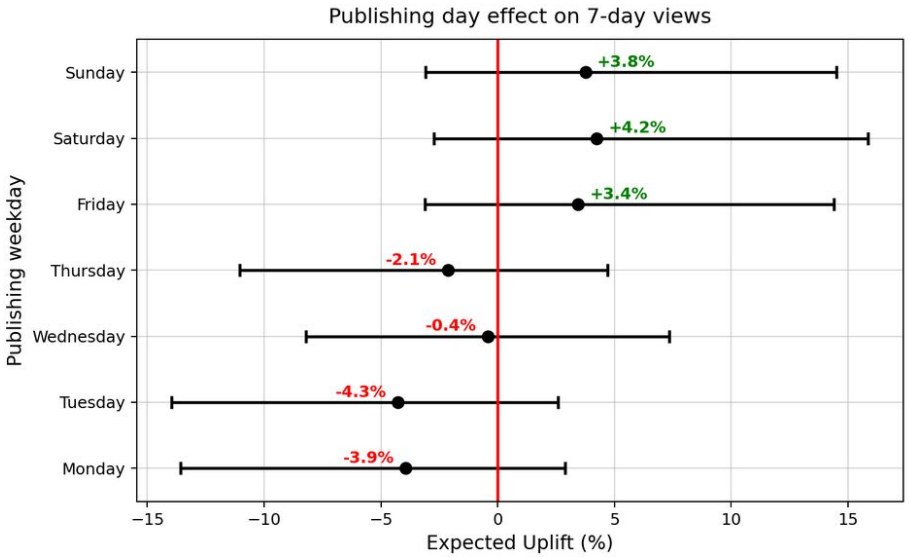
$$y_i \sim \mathcal{N}(\mu_i, \sigma_{obs})$$

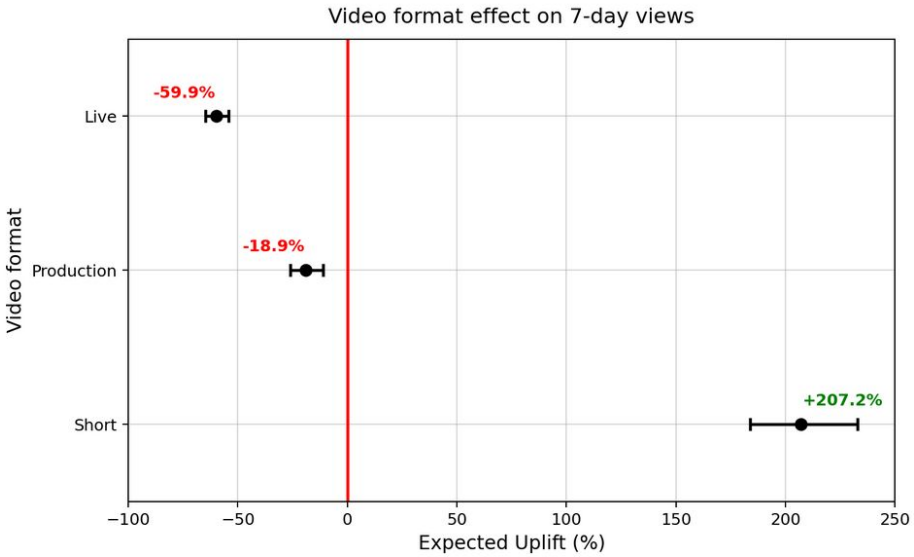
$$\mu_i = \alpha + \delta_{d[i]} + \phi_{f[i]}$$

**Effect of publishing weekday**

**Video format effect**

# Insight 1: Results




| comparison | median_uplift_pct | low_95_pct | high_95_pct | prob_gt_zero |
|---|---|---|---|---|
| Saturday vs Monday | 8.772961 | -2.530052 | 29.050609 | 0.90225 |

Conclusions

- Publishing timing has a modest impact. Weekend uploads generate **~10%** more views than Monday.
- Video format is the dominant driver of performance. Shorts generate **about 3×** more views than other formats.

# Insight 2: Content Lifetime by Video Type and Platform

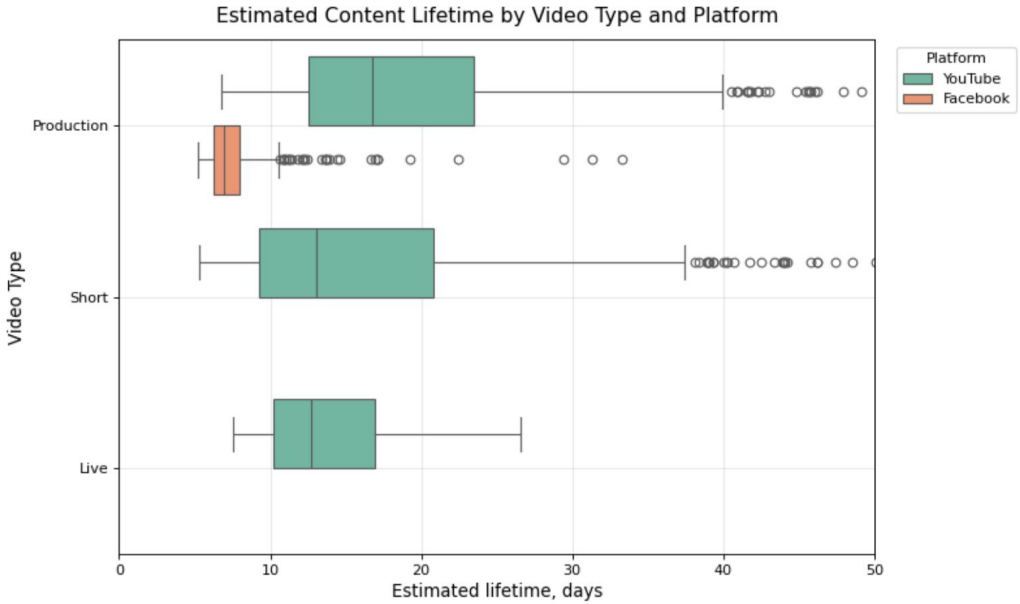
## View Growth Model

τ - Content Lifetime

$$V(t) = V_{\mathrm{SAT}}\left(1 - e^{-t/\tau}\right)$$

## Data Source

- Monthly cohort data tracking video views after publication

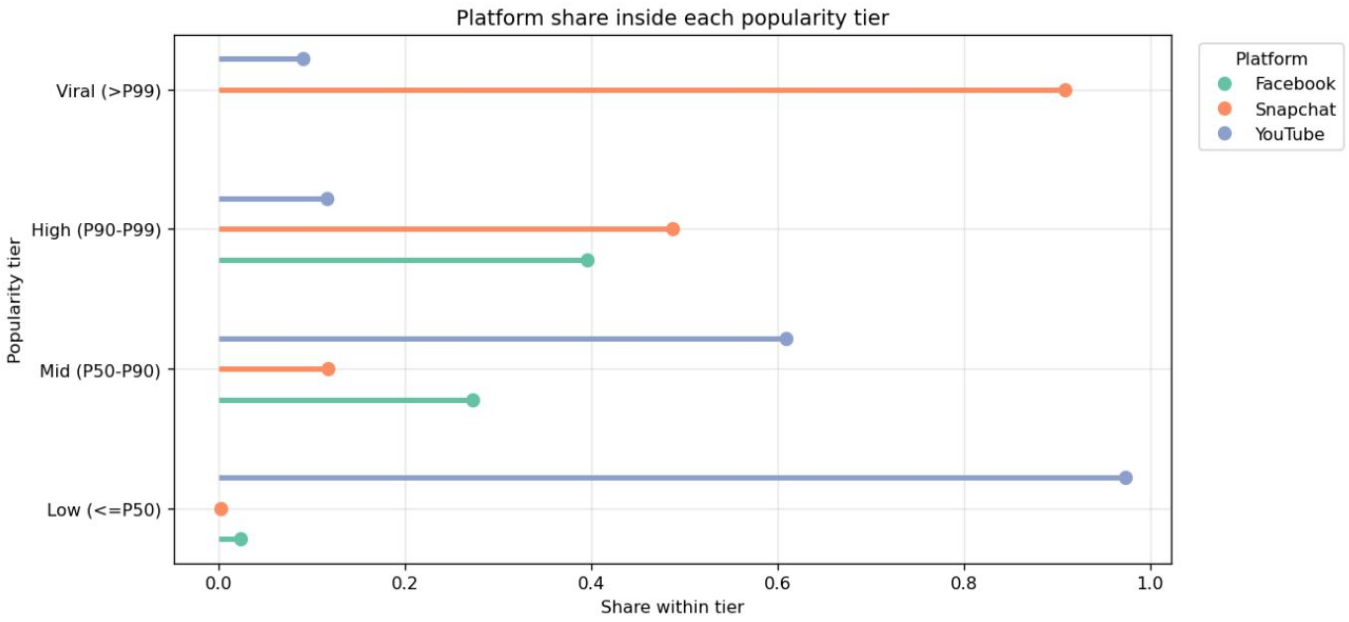



Content on Facebook seems to “burn out” much faster than on YouTube. Typical lifetime is ~6–7 days vs ~15–20 days.

# Insight 3: Platform Fingerprint of Viral Content

**Data Source**

• Cross-platform dataset tracking original videos and their reposts across platforms




Viral content is overwhelmingly driven by Snapchat, while YouTube dominates the long tail of low-performing videos.

# What I would do with more time

- Connect insights directly to business impact and revenue.
- Test additional model assumptions and interactions where it matters.
- Use richer signals (video, comments, text) and modern deep learning to extract deeper patterns.